IUPAC naming

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator. Pick what your course covers once, and IUPAC naming, Synthesis and future generators all use it, unless you customize a generator separately.

Allowed structures and functional groups

Select all Clear

Generators only build molecules from what's checked.

Hydrocarbons

- [x] Alkane
- [x] Alkene
- [x] Alkyne
- [ ] Aromatic

Functional groups

Core

- [x] Alcohol
- [x] Alkyl halide
- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

Carboxylic acid derivatives

- [ ] Carboxylic acid
- [ ] Acid chloride
- [ ] Anhydride
- [ ] Ester
- [ ] Amide
- [ ] Nitrile

Auxiliary

- [ ] Amine
- [ ] Nitro

Molecule size

Number of carbons in each molecule. 3 to 10

Include rings

Cycloalkanes and other cyclic molecules.

Include heteroatoms in rings

Pyridine, furan, thiophene and similar. Needs Aromatic checked.

Stereochemistry

When included, answers must give correct R/S and E/Z. Off | Include

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

IUPAC naming

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

MOLECULE OPTIONS

Use the shared Molecules settings
Semester 1 · 7 groups · 3–10 carbons · rings on · stereo off · Edit shared

Customize for IUPAC naming only
Changes here won't affect other generators.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents.
3 to 10

Most senior functional group
Chosen from the groups allowed under Molecule options.
Any allowed

Number of principal groups
1 to 2

Number of secondary groups
0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

IUPAC naming

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming !

Synthesis

Acid/Base Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents.
4 to 3

! Minimum is larger than maximum.

Most senior functional group
Chosen from the groups allowed under Molecule options.
Carboxylic acid

! Carboxylic acid isn't in your allowed groups. Allow it

Number of principal groups
0 to 2

Number of secondary groups
0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

IUPAC naming

Settings

ALL GENERATORS

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GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Before you start: choose what to practice

Pick what your course covers. Shared by every generator; change it any time with Settings.

Allowed structures and functional groups

Select all Clear

Generators only build molecules from what's checked.

Hydrocarbons

- [x] Alkane
- [x] Alkene
- [x] Alkyne
- [ ] Aromatic

Functional groups

Core

- [x] Alcohol
- [x] Alkyl halide
- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

Carboxylic acid derivatives

- [ ] Carboxylic acid
- [ ] Acid chloride
- [ ] Anhydride
- [ ] Ester
- [ ] Amide
- [ ] Nitrile

Auxiliary

- [ ] Amine
- [ ] Nitro

Molecule size

Number of carbons in each molecule.

3 to 10

Include rings

Cycloalkanes and other cyclic molecules.

Use defaults

Start practicing →

IUPAC naming

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator.

Advanced options

	CARBONS		AMINES (needs Amine checked)	
Secondary (2°) Carbon bonded to 2 carbons · R_2NH	0 to 4		0 to 1	
Tertiary (3°) Carbon bonded to 3 carbons · R_3N	0 to 2		0 to 1	

Ring topology
Single rings are always included when Include rings is on.
☐ Spiro ☐ Fused ☐ Bridged

Aromatic ring positions
Substituent patterns on disubstituted benzene rings. Needs Aromatic checked.
☐ Ortho (1,2) ☐ Meta (1,3) ☐ Para (1,4)

Functional group count
Total number of functional groups in each molecule.
0 to 3

Functional group distance
How far apart two functional groups are (Warren, *The Disconnection Approach*).
☑ 1,1 ☑ 1,2 ☑ 1,3 ☑ 1,4 ☑ 1,5

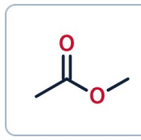
Custom fragments
Every generated molecule must contain all of these fragments.

+ Add fragment



C1CCCCC1

Edit Remove



C(=O)OC

Edit Remove

! Contains an ester, which isn't checked above. Allow ester

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

IUPAC naming

Settings

ALL GENERATORS

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Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator.

Add a custom fragment

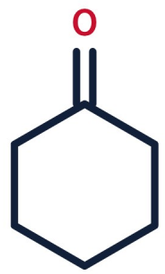
Draw | Enter SMILES

SMILES

O=C1CCCCC1

One fragment per entry, e.g. C1CCCCC1 for a cyclohexane ring.

PREVIEW



Cyclohexanone fragment · Open in Draw to edit

Cancel | Add fragment

eeds Amine checked)

to 1

to 1

Fused Bridged

Meta (1,3) Para (1,4)

0 to 3

1,3 1,4 1,5

+ Add fragment

t checked above.

Allow ester

Changes save automatically and apply to your next problem

Reset this section to defaults

Settings

ALL GENERATORS

Molecules >

GENERATORS

IUPAC naming >

Synthesis Laptop or tablet >

Acid/Base Coming soon

Settings

IUPAC naming

MOLECULE OPTIONS

- (•) **Use shared Molecules**
  Semester 1 · 7 groups · Edit shared
- ( ) **Customize for IUPAC only**

NAMING PROBLEMS

Parent chain length	3	to	10
Most senior group Any allowed	Edit		
Principal groups	1	to	2
Secondary groups	0	to	2

PRACTICE

Count skipped problems

✓ Saves automatically

Reset section